OPEN AGENT HACKATHON 2026 · TRACK 1 — SOLVING FRAGMENTED INTELLIGENCE

RecallWhy

Agents that tell a retailer which recalled products are on its shelves, which lots, which stores, how many customers to call — **and why they believe it.**

Zetaris federated SQL · Meterless H-MEM + World Model + Markovian · Meterless Nightshift · NVIDIA Nemotron 3 Ultra · real products from USDA FoodData Central and Open Food Facts

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github.com/mikey92/recallwhy

1 · PROBLEM AND USERS

515 federal recall notices since July. Each one is a manual cross-check.

A regional retailer's **product-safety / QA manager** reads every CPSC and FDA notice and must answer:

- Do we sell this — or something that only looks like it?
- Which lots? Which stores still hold them?
- How many customers bought it since the recall began?
- Who supplied it?

Dec 2025: FDA warning letters to Kroger, Albertsons, Target and Walmart — recalled infant formula still on shelves in 175+ locations across 36 states, up to three weeks. Staff cited "confusion regarding which specific lots were affected."

184 + 333

CPSC recalls and FDA food enforcement reports, 1 Jul – 21 Oct 2026 (snapshot of 22 Oct in `sample/`; RecallWhy reads the same feeds live)

The answers live in four systems that were never meant to meet — two **recall feeds**, the **ERP** and the **supplier portal** — each writing a product its own way: UPC-12 in the ERP, GTIN-14 from a third of suppliers, digits buried in notice text. A wrong "not ours" leaves a contaminated lot on the shelf; a wrong "ours" pulls good stock.

2 · SOLUTION

Five agents map the fragments, then investigate every recall and return a verdict with its evidence.

Cartographer

Lists every Zetaris source and column, profiles identifiers, tests cross-source joins. Finds the GTIN-14 suppliers upfront. When the Auditor hits rows the key can't reach, it diagnoses that supplier, tests the fix on all its ship notices, and corrects the map.

Scout

Reads both live feeds and the item master through Zetaris; one federated UPC sweep; opens a case per recall that touches the catalog.

Investigator

UPC / model rules, Nemotron judge for text matches, lot codes, prior decisions from H-MEM, exposure by store and lot with federated SQL.

Auditor

Can block. Re-checks the live notice, quotes and lots verbatim, re-derives every number with its own SQL, checks each brief sentence. ≤2 revisions, then a human.

Briefer

Writes only from the World Model context: numbered evidence, each with the SQL or model call behind it. Then the action packet and the weekly digest.

ACT NOW · NOTIFY · NO EXPOSURE · NOT OURS · NEEDS HUMAN

3 · ARCHITECTURE

Every agent reads data only through Zetaris.

CPSC Recall API
REST · JSON array

openFDA enforcement
REST · JSON

ERP Postgres
JDBC · real food items, synthetic stock

Supplier portal
JDBC · synthetic

→

Zetaris Lightning SQL
REST tables · schemastore views
JDBC datasources · federated joins

→

Cartographer → Scout → Investigator ⇄ Auditor → Briefer

Auditor → Cartographer: barcode rows the map can't reach · Cartographer → Auditor: corrected map

Meterless **H-MEM**
every decision, trust 0.6 agent / 0.95 human

Meterless **World Model**
facts + provenance → ContextBuilder

Meterless **Nightshift** → NVIDIA **Nemotron 3 Ultra 550B**
one injected generator · judge · lot reader · why-paragraph · routed model + receipt on every call

Meterless **Markovian**
weekly digest over every case, bounded chunks + carryover

action packets · report.html · trace.jsonl · review queue

4 · STACK

Small, typed, runnable from a clean clone.

LAYER	WHAT
Data access	Zetaris Lightning SQL over its REST API (`/api/proxy/sql-editor/sqls/run-query`)
Agents	TypeScript on Node 22, no agent framework — five agents, one orchestrator, one trace
Memory, state, long reads	Meterless H-MEM, World Model and Markovian reference engines (Apache-2.0), mounted in `src/engines/`
Model	Meterless **Nightshift** Runtime API (OpenAI-compatible) → NVIDIA **Nemotron 3 Ultra 550B-A55B**; no model and a human queue when Nightshift is absent
Retailer data	Postgres 16 in Docker — food items are real products (USDA FoodData Central, Open Food Facts: 95 recalled, 86 same-brand, 260 more); stock, sales and ship notices synthetic
Entry points	`POST /run` · `POST /approve` API · Dockerfile · CLI · live trace viewer · review screen
No-credential mode	`SAMPLE_MODE=true` : same agents and SQL on an in-process DuckDB stand-in, labelled SAMPLE
Development	Cursor (agent scaffolding, generator, SQL layer, tests) · 79 tests incl. Meterless engine tests

5 · HOW ZETARIS AND METERLESS ARE USED

Load-bearing, not bolted on.

Zetaris

- Cartographer maps sources with Zetaris's catalog: `SHOW DATASOURCES`, `DESCRIBE DATASOURCE TABLE`, `SHOW LIGHTNING DATABASES` → profiles and join tests
- 2 live Lightning REST tables → schemastore views
- 2 JDBC datasources (ERP, supplier portal)
- Scout: **REST × JDBC** join — live FDA notice digits ⋈ ERP UPCs
- Investigator: ship notices ⋈ suppliers ⋈ ERP item master
- Auditor: re-queries the live notice and re-aggregates every number itself
- Setup is code: `npm run setup:zetaris`

Meterless

- **H-MEM**: every adjudication stored with trust; human decisions (0.95) are followed on the next run; audited agent decisions are reused without a model call; trust ledger logs reads/writes
- **World Model**: recalls, cases, findings as facts; every write carries provenance (agent, SQL id, model call)
- **ContextBuilder** is the Briefer's only input — it cannot cite what the team didn't establish
- **Markovian**: the weekly digest reads every case of a run in bounded chunks with a compressed carryover (≤256 tokens)
- **Nightshift** is the primary AI provider: every model call, with its receipt and routed model in the trace

Map the fragments first. Then trust no one — including the model.

- **The map is a hypothesis the Auditor can send back.** Rows the key can't reach go to the Cartographer, which learns the supplier's rule (padding, dropped check digit) and proves it on all that supplier's ship notices.
- **The model proposes, SQL proves.** For a format the code has no rule for (a 10-digit core), Nemotron 3 Ultra proposes a transform from sampled pairs in a fixed op set; it is adopted only if it holds on every one of the supplier's rows (193 to 390 rows, four suppliers), then written back into Zetaris as a view, read back, and checked by Zetaris's own lineage and data-quality rule (every row valid). Judge off: **22** of 194 caught with known rules, **103** with proposals, **11** without the Cartographer.
- **Re-derives every number in Zetaris** with differently written SQL; a mismatch blocks the brief.
- **Re-reads the live notice** (REST table) to confirm a UPC match, instead of trusting the parser.
- **"Cheese ≠ pepperoni"**: a quote can be copied from the notice and still describe another variety — every product word of our item must appear in the notice. Live run: blocked, re-judged, ruled *not ours*.
- **Lot codes the model reads must appear verbatim** in `code_info`; invented lots are dropped.
- **Brief sentences are checked** against the evidence they cite; a draft that fails the check is rewritten or replaced by the template.
- **Human decisions change the next run** through H-MEM, and the brief cites them.

fda:H-1396-2026 · GIAS FOOD INC. · 2026-09-03

> Lemon Alfredo Fettuccine packed in a yellow plastic sealed bag with pink and black lettering; label reads in part as...

WHY

The recall notice identifies GIAS FOOD INC. Lemon Alfredo Fettuccine with UPC 194346442706 due to contamination with Listeria monocytogenes E1 E2 E3 E7. Our inventory matches this UPC and holds the specific lots L6079C and L6080C explicitly listed in the notice E8 E9. We currently have 130 units on hand across 4 stores and have sold 102 units to 47 loyalty customers since the recall date of 2026-09-03 E4 E11 E12 E13 E14.

WHAT WE MATCHED

- NW-100069 Lemon Alfredo Fettuccine — **same** (rule, upc): our UPC 194346442706 is printed in the notice

ACTIONS

- Pull lots L6079C, L6080C from 4 store(s): S012 (64), S023 (43), S014 (19), S003 (4) E12
- Send recall notice to loyalty customers who bought affected lots since 2026-09-03: at most 47 E14
- Contact Supplier S028 (synthetic) for credit and replacement E15

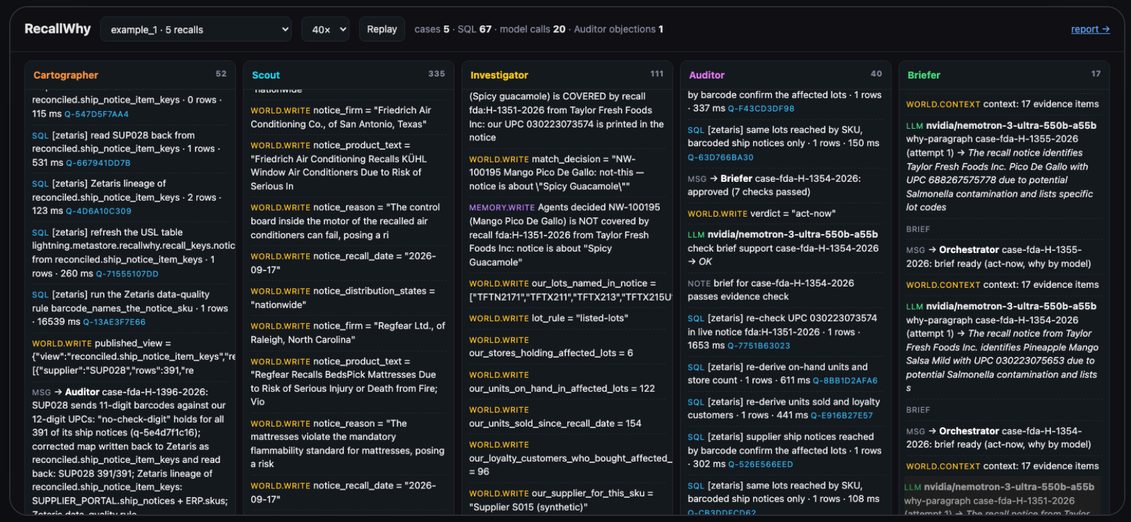
CAVEATS

- Retailer stock, sales, lots and ship notices are synthetic (Northwind Mercantile, generated for this demo); food items are real products with real barcodes (USDA FoodData Central, Open Food Facts).
- Customer count sums weekly distinct counts, so it is an upper bound.

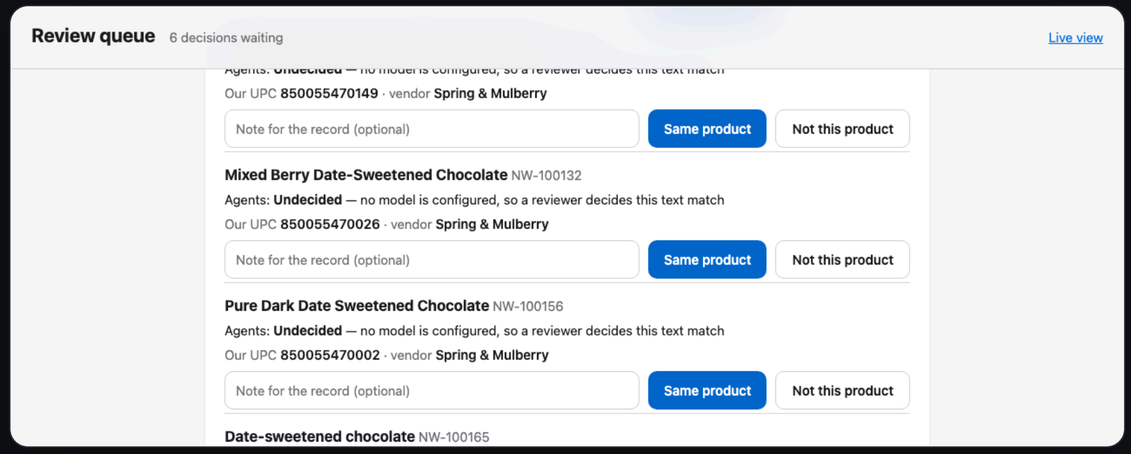
EVIDENCE

- E1 notice_firm: "GIAS FOOD INC." — *zetaris q-11d60a4f83 by Scout*
- E2 notice_product_text: "Lemon Alfredo Fettuccine packed in a yellow plastic sealed bag with pink and black lettering; label reads in part as \"bettergoods Lemon Alfredo Fettuccine, MA — *zetaris q-11d60a4f83 by Scout*
- E3 notice_reason: "Contamination with Listeria monocytogenes" — *zetaris q-11d60a4f83 by Scout*
- E4 notice_recall_date: "2026-09-03" — *zetaris q-11d60a4f83 by Scout*
- E5 notice_distribution_states: "nationwide" — *zetaris q-11d60a4f83 by Scout*
- E6 notice_code_info: "L6080C SEP 20 2027 L6079C SEP 19 2027" — *zetaris q-11d60a4f83 by Scout*
- E7 notice_upcs: ["194346442706"] — *zetaris q-11d60a4f83 by Scout*
- E8 match_decision: "NW-100069 Lemon Alfredo Fettuccine: same — our UPC 194346442706 is printed in the notice" — *rule by Investigator*
- E9 our_lots_named_in_notice: ["L6079C","L6080C"] — *zetaris q-db1c3095f6 by Investigator*
- E10 lot_rule: "listed-lots" — *agent by Investigator*
- E11 our_stores_holding_affected_lots: 4 — *zetaris q-db1c3095f6 by Investigator*
- E12 our_units_on_hand_in_affected_lots: 130 — *zetaris q-db1c3095f6 by Investigator*
- E13 our_units_sold_since_recall_date: 102 — *zetaris q-227f7d85c3 by Investigator*
- E14 our_loyalty_customers_who_bought_affected_lots: 47 — *zetaris q-227f7d85c3 by Investigator*
- E15 our_supplier_for_this_sku: "Supplier S028 (synthetic)" — *zetaris q-b5523d1bc8 by Investigator*
- E16 our_stores_with_units_on_hand: ["S012 MN: 64","S023 GA: 43","S014 WI: 19","S003 OR: 4"] — *zetaris q-db1c3095f6 by Investigator*
- E17 verdict: "act-now" — *agent by Auditor*

Each case opens with what the agents did (here: a supplier rule proven by SQL on 391/391 and written back to Zetaris), then the cited brief and the trace.



Live viewer: the agents' events as they happen, from the trace file.



Review screen: the open case, the Auditor's objection, and a decision stored in Meterless H-MEM.

7B · THE WORK IT HANDS ON

Every recall with our stock ends in a packet a person approves.

pull-list.csv

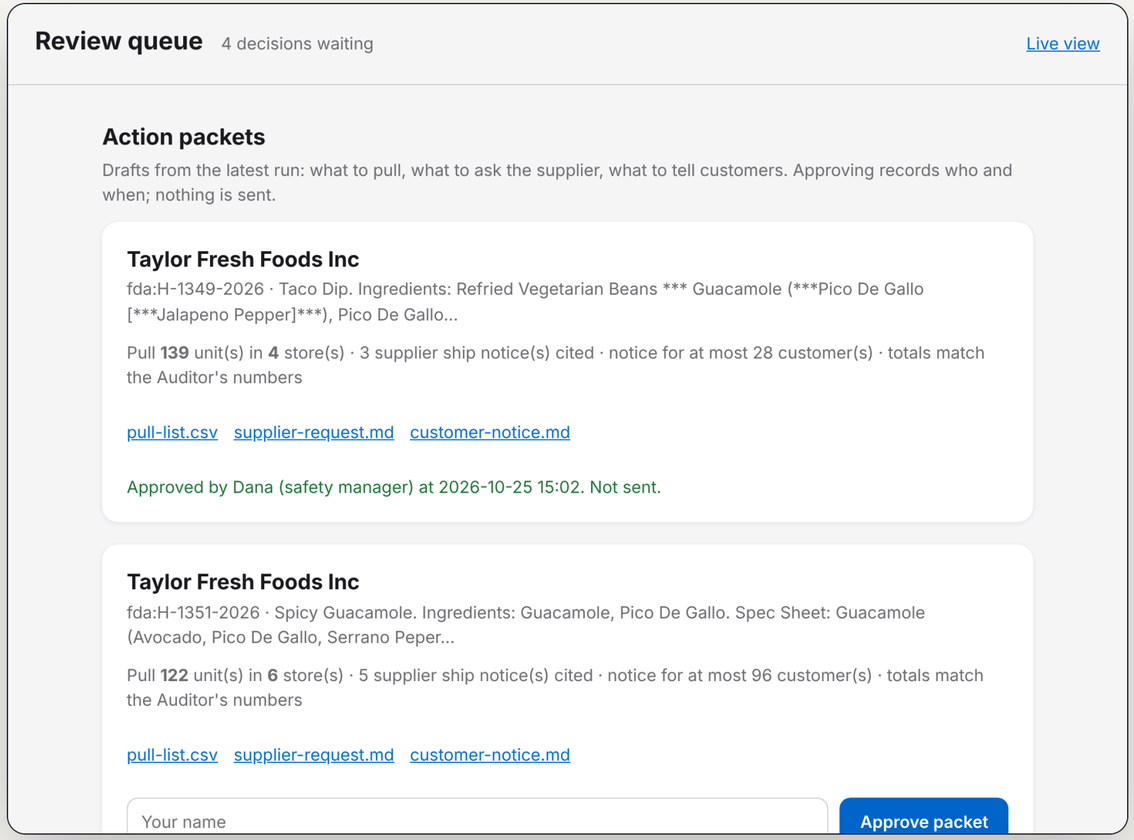
One row per store and lot: store, city, item, UPC, lot, units. **Withheld unless it adds up to the on-hand number the Auditor re-derived with its own SQL.**

supplier-request.md

Credit and replacement request citing the supplier's own ship notices (ASN, lot, date, units), one per supplier.

customer-notice.md

For loyalty customers who bought affected lots — a count; the loyalty system resolves who. Reason quoted from the notice.



Review screen: drafts from the latest run, approved with a name and a time. Nothing is sent. Plus `digest.md`: what needs the manager first, read through Markovian.

8 · RESULTS

134 of 146 recalled items caught, 1 missed, 1 false alarm — on real products.

SETUP (322 RECALL/SKU PAIRS)	CAUGHT (OF 146)	WRONGLY DISMISSED	FALSE ALARMS (OF 176)	TO HUMAN
No Cartographer — rules only	53	0	0	269
Cartographer, known rules only	89	0	0	233
Cartographer + model-proposed transforms (no judge)	99	0	0	223
+ NVIDIA Nemotron 3 Ultra via Nightshift	**134**	**1**	**1**	**30**

TEXT MATCHES (47 COVERED OF 223)	CAUGHT	WRONGLY DISMISSED	FALSE ALARMS	TO HUMAN
Ultra: judge's first answer	43	4	2	0
Ultra: after the Auditor loop	**45**	**1**	**1**	**20**

292 of 322 pairs (91%) settled without a person, 290 correctly; 30 go to a reviewer with the reason. 0 of 392 model calls failed. On the real products, every recall whose barcode we carry is found by rule (97 of 97), with 0 of 152 false matches.

Real Investigator ⇄ Auditor loop on every candidate pair from 515 recalls (sample-mode data). Food items are real products (USDA FoodData Central, Open Food Facts); stock, sales and ship notices are synthetic. Details: eval/EVAL.md, eval/REAL.md.

9 · NEXT

From demo to a safety desk.

- More feeds through Zetaris: USDA FSIS, NHTSA equipment (car seats), Health Canada
- Lot capture from receiving scans instead of the item master
- Send the approved packet: loyalty system for customers, EDI for the supplier credit, store task system for the pull
- Cartographer onboarding new feeds on its own: catalog → profile → join test
- Real stock and ship notices from a pilot retailer (the food catalog is already real)
- Review screen for a team: assignment, and each decision's history from the H-MEM trust ledger

Limits today: stock, sales and ship notices are synthetic (the food items are real); non-food titles come from the notice; proposed transforms are limited to string ops; lexical Auditor checks; CPSC rarely prints UPCs; hosted model rate limits make a full evaluation take over an hour.